

OAK RIDGE National Laboratory | MANUFACTURING DEMONSTRATION FACILITY

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Adamantine Advanced Tutorials

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Today’s Tutorials

- Exercise 02: Set boundary conditions and time step size
- Exercise 03: Define the heat source
- Exercise 04: Set the material properties
- Exercise 05: Run ensemble simulation
- Exercise 06: Use data assimilation to improve solution

Exercise 2: Boundary and time stepping

- Boundary:
  - Thermal boundary condition supported are: adiabatic, radiative, and convective
  - A boundary can be both radiative and convective
  - A general boundary can be applied to the entire domain boundary
  - One can overwrite the boundary condition for any boundary ids
- Time stepping:
  - Set the time step size
  - Set the end time of the simulation or until the scan path ends

Exercise 2: Boundary and time stepping

- Exercise is in the ***work/02_Adamantine/02*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#boundary-required and https://adamantine-sim.github.io/adamantine/doc/input_file.html#time_stepping-required
- Run adamantine using: *adamantine –i input.info*
- The solution of the exercise is given in the solution.info file

Exercise 3: Heat Source

- Adamantine supports an arbitrary number of heat sources but the heat sources must be in the same z-plane
- Two types of sources are supported: goldak and electron beam
- Sources must be enumerated consecutively starting at zero
- Two scan path formats are supported: segment format and event format
- Segment format is similar to the format used by AdditiveFoam but the power coefficient multiplies the maximum power set in the input file

Exercise 3: Heat source

- Exercise is in the ***work/02_Adamantine/03*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#sources-required
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file

Exercise 4: Material Properties

- Materials properties can be set for powder, solid, and liquid state
- Adamantine solves the anisotropic heat equation → the thermal conductivity needs to be given for x, y, and z (follows deposition axis)
- It is not necessary to set all the material properties
- Material properties can be given as a polynomial of the temperature or as a table
- Important:
    - Define a material for each material id in the mesh
    - Material ids are consecutive numbers starting at 0

Exercise 4: Material Properties

- Exercise is in the ***work/02_Adamantine/04*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#materials-required
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file

Exercise 5: Ensemble simulation

- Adamantine can run multiple simulations with different input values described by a normal distribution
- Standard deviations must be set in the ensemble section
- Mean values are set in the regular sections
- This functionality is applicable only to scalar variables
- This does not work for temperature-dependent variables
- The values picked can be found in *prefix_m0_data.txt, prefix_m1_data.txt, etc.*

Exercise 5: Ensemble simulation

- Exercise is in ***work/02_Adamantine/05*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#ensemble-optional
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file

Exercise 6: Data assimilation

- Data assimilation can improve the accuracy of simulation
- It requires calibrated experimental data
- Simulation errors is computed using ensemble simulations
- EnKF introduces spurious long-range correlations between distant points
- Localization reduces these correlations using a cutoff radius

Exercise 6: Data Assimilation

- Exercise is in ***work/02_Adamantine/06*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#data_assimilation-optional and https://adamantine-sim.github.io/adamantine/doc/input_file.html#experiment-optional
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file

Exercise 7: Drip feed scan paths

- To support closed-loop control, Adamantine can wait on the scan path to update
- Updating the scan path needs to be an atomic operation
    - Appending to a file is not atomic
    - You need to create a temporary file and then rename it using *mv*
- Make sure that the file name is correct

Exercise 7: Drip feed scan paths

- Exercise is in ***work/02_Adamantine/07*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#time_stepping-required and https://adamantine-sim.github.io/adamantine/doc/scan_path.html#event-format
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file

Exercise 8: Postprocessor, refinement, and units

- Postprocessor:
  - Set the prefix of output files
  - Set the output frequency
- Refinement
  - Set the maximum number of time a cell can be refined
  - Set the number of time steps between refinement
- Units
  - Set units for the mesh and the heat source

Exercise 8: Postprocessor, refinement, and units

- Exercise is in ***work/02_Adamantine/08*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#post_processor-required, https://adamantine-sim.github.io/adamantine/doc/input_file.html#refinement-required, and https://adamantine-sim.github.io/adamantine/doc/input_file.html#units-optional
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file